Alexander Jando

Hamilton, ON | West Vancouver, BC

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Education

McMaster University **Expected April 2030**
Mechatronics Engineering and Management Co-op (B.Eng.Mgt.) *Hamilton, ON*

Technical Skills

Programming: Python, C/C++, Java, JavaScript/TypeScript, MATLAB, SQL
Hardware & Testing: ESP32, Arduino, Raspberry Pi, oscilloscopes, firmware flashing, motor control
CAD & Prototyping: Fusion 360, Autodesk Inventor, Onshape, 3D printing, laser cutting
Software & Tools: Pytest, Playwright, Jenkins, OpenCV, Flask, React, Git, Linux, Jira

Engineering Experience

Corinex Communications **Vancouver, BC**
Hardware R&D Engineering Co-op *June – August 2026*

- Built a test rack supporting 192 concurrent devices on a virtual network and flashed embedded firmware for testing and validation.
- Developed software to automate bill of materials (BOM) exports and collect related engineering drawings and specification sheets into folders, saving the hardware team an estimated 30 minutes per day.
- Designed and built an in-house pressurized IP67 test setup, custom fixtures, and workshop tools for product assembly, testing, and prototyping.
- Tested and tuned communication hardware with oscilloscopes to analyze frequency bands, investigate hardware behavior, and verify system performance.
- Designed components in Fusion 360 and iterated 3D-printed prototypes for production and workshop use.

Corinex Communications **Vancouver, BC**
Software QA Engineering Co-op *May – June 2026*

- Developed automated tests with Python, Pytest, and Playwright and integrated them into Jenkins CI for repeatable software validation.
- Investigated defects across software and embedded hardware with QA and firmware engineers; verified fixes.

Hypercharge Networks **North Vancouver, BC**
EV Charger Technician Intern *June – August 2023*

- Prepared and configured 100+ EV chargers by initializing firmware, registering units on the network, and checking connectivity and performance before deployment.
- Diagnosed faulty or uncalibrated units, replaced internal components, and documented hardware tests and quality checks in a production environment.

Engineering Projects

Robotic Order-Packing System *Project Manager & Software Lead*

- Led a five-person team building a Q-Arm packing system for McMaster ENG 1P13; managed scope and schedules.
- Designed a 4:1 rack-and-pinion gripper in Autodesk Inventor within a 100 g limit; iterated 3D-printed prototypes to resolve gear meshing, tolerance, and rigidity issues.
- Developed modular Python control software for product lookup, arm motion sequencing, packing, and operation logging.

Accessible Macro Pad *Project Coordinator & Hardware Designer*
Accessible Macro Pad ↗

- Designed a Bluetooth device with three large controls and six configurable mobile actions; created the enclosure, keycaps, and ESP32 resistor-ladder circuit.
- Contributed to firmware and the React Native app; repaired connections and calibrated ADC thresholds for reliable detection of all three inputs.

FIRA RoboWorld Cup Autonomous Race Car *Computer Vision & Controls Developer*

- Built an autonomous car with Raspberry Pi, Arduino, Python, and OpenCV; implemented edge detection and PID steering. Placed 1st nationally and 2nd internationally in the autonomous car competition.

NoNotes – AI Note-Taking Platform *Backend Development Lead*

- Led Flask and SQLite backend development, integrating speech-to-text, NLTK, and summarization to create searchable lecture notes; won 1st place at HappyHacks II in 48 hours.

Languages

English (Native) | French (Fluent, DELF B2)